

Shantanu Shastri

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EDUCATION

Carnegie Mellon University

Master of Information Systems Management,

December 2026

GPA: 3.85

Manipal Institute of Technology

Bachelor of Electronics and Communications Engineering with Minor in Data Science,

August 2022

GPA: 3.99

Relevant Coursework: Data Structure & Algorithms, Java, Python, Object Oriented Programming, Machine Learning, Linear Algebra

SKILLS

Languages & Libraries: Java, C++, JavaScript, Python, TypeScript, SQL, NumPy, Pandas, Bash/Shell, PyTorch

Developer Tools/Frameworks: Spring Boot, gRPC API, System Design, Node.js, SQL Databases, SonarQube, Veracode, Git, JUnit, Postman, Angular, TensorFlow, Deep Learning, NLP, Linux, Grafana, Redux, Flask, React.js, FastAPI, MCP, A2A, RAG, Grafana, Prometheus

Infra & Database: AWS, Azure, Docker, Kubernetes, Kafka, MySQL, MongoDB, Redis, Neo4j, Azure Blob, Milvus, MLflow

EXPERIENCE

BlackRock

New York

AI Engineer Internship

June 2026 – August 2026

- **Multi-agent orchestration platform:** Built a production ETL & multi-service backend from Sybase using FastAPI, Neo4j, Milvus, PostgreSQL, and Kafka, with a live-polling pipeline for real-time ingestion into a knowledge graph.
- **Self-evolving vector graph:** Architected search pipeline (Milvus, Neo4j) with closed-loop decision capture, combining semantic embeddings with structured graph traversal and LLM-based action classification.
- **Evaluation & Policy Optimization:** Created offline-online evaluation frameworks that track agent latency, confidence scores, data drift, and provenance, and use policy optimization to improve agent accuracy, handling 3.5M records and \$14T in cash.

Software Developer 2 - Associate

January 2025 – July 2025

- **System Scalability & Optimization:** Architected a central data framework and TTL Azure Blob microservices to resolve legacy bottlenecks, successfully **automating 1.5M+ touchpoints** across 117 clients while significantly reducing infrastructure storage costs.
- **gRPC API & Caching:** Engineered a distributed read-through Ignite cache to address high-latency APIs, cutting response times by 3x.

Software Developer 1 - Analyst

July 2022 – December 2024

- **Database Optimization:** Achieved **37% reduction in query latency** by refactoring critical legacy database tables and leveraging composite indexes for faster data retrieval, significantly improving user experiences for key workflows.
- **Full-stack Development:** Spearheaded end-to-end development of **responsive, high-performance Angular-based UI** using NgRx redux state management and integrated **Azure's LLM APIs** to deliver **real-time AI** capabilities directly into the user workflow.

Multi-Instance – Multi-Threading

- **Distributed Systems Architecture:** Resolved critical singleton bottlenecks by **engineering a sharded, multi-instance server** environment with **Zookeeper leader election**, effectively distributing load and **boosting system performance by 24%**.
- **DevOps & Quality Engineering:** Built comprehensive **CI/CD pipelines** using **Azure DevOps**, with integrated **JUnit/Jacoco** testing suites, **reducing time-to-market by 40%** while maintaining **SONAR scores >90%**.

Software Developer Internship

January 2022 – June 2022

- **Application Modernization:** Re-architected legacy Pearl/Tomcat applications into a **modular Angular & Java REST environment**, implementing **micro-frontends** to increase code reusability across UI domains significantly.

PROJECTS

Zulip Open-Source LLM Extensions

January 2026

- **LLM Feature Engineering:** Extended the Zulip platform by building a "Message Recap" engine using **Tornado, Redis and RabbitMQ** that leverages **Large Language Models (LLMs)** to synthesize unread streams into concise, clickable summaries.
- **Cost & System Optimization:** Designed a cost-efficient **Topic Drift Detection system** to correct conversation titles in real-time, optimizing interaction models to significantly minimize token usage costs.

Production Recommendation & MLOps System

March 2026

- **Hybrid Recommender & Serving:** Developed hybrid recommender using LightFM (WARP) and SVD++ (722 recs/s throughput) with Llama-3-based cold-start logic, containerizing the service with Docker and a deterministic SHA-256 load balancer
- **Automated MLOps & Auditing:** Built a 4-hour retraining loop with zero-downtime model reloading tracked via MLflow, monitored metrics using Prometheus/Grafana, and performed STRIDE, STPA, and demographic fairness audits.